AI-powered Legal Assistant for Indian Criminal Law

PROBLEM STATEMENT

India's criminal justice system, currently transitioning from the IPC to the Bharatiya Nyaya Sanhita (BNS), remains largely inaccessible to the public due to its dense, technical language and complex statutes. Citizens often struggle to navigate legal rights, bail eligibility, and specific provisions, as traditional search systems lack the contextual intelligence needed to bridge this information gap. This project addresses these barriers by developing an AI-driven guidance system that translates complex criminal law into structured, context-aware, and simplified insights for the average citizen.

PROJECT OVERVIEW

The AI-Based Legal Intelligence Agent is a RAG-powered web platform that simplifies Indian criminal law by analyzing natural language case descriptions. By retrieving relevant sections from the IPC, BNS, and CrPC, the system provides grounded, context-aware guidance on punishments and bail eligibility. Ultimately, the project enhances legal literacy through semantic search while maintaining the necessity of professional legal counsel.

SOLUTION OFFERED

The proposed solution is an AI-powered legal agent that utilizes **Natural Language Processing (NLP)** and **vector embeddings** to interpret case descriptions meaningfully. Operating on a **Retrieval-Augmented Generation (RAG)** framework, the system converts user input into high-dimensional vectors via sentence transformers, performs a semantic similarity search using a **FAISS-based vector database**, and retrieves the most relevant legal sections. By grounding a language model in this retrieved context, the system generates structured, accurate responses regarding applicable sections, punishment details, and bail eligibility while minimizing AI hallucinations. This scalable architecture moves beyond rigid keyword searches, successfully bridging the gap between complex statutes and accessible legal intelligence.

END USERS

Common citizens, Law students, Researchers, Junior practitioners, Legal aid organizations

TECHNOLOGY USED

Backend & API Management: FastAPI (Python) for building a high-performance, asynchronous API and Python Test Scripts to validate LLM integration and response accuracy.
Intelligence & Local Inference: Ollama as a local LLM runtime for private, offline response generation and RAG (Retrieval-Augmented Generation) architecture to ground answers in verified legal statutes.
Semantic Search & Vectorization: Sentence Embedding Models for converting case text into dense vectors and a Local Vector Index (FAISS) for efficient, context-aware similarity searching of legal provisions.
Data Architecture: A Structured Legal Corpus containing IPC, BNS, and CrPC sections, converted into high-dimensional embeddings and stored locally for rapid retrieval.
Frontend & User Interface: HTML, CSS, and JavaScript to provide a clean, responsive, and intuitive web interface for natural language user interaction.